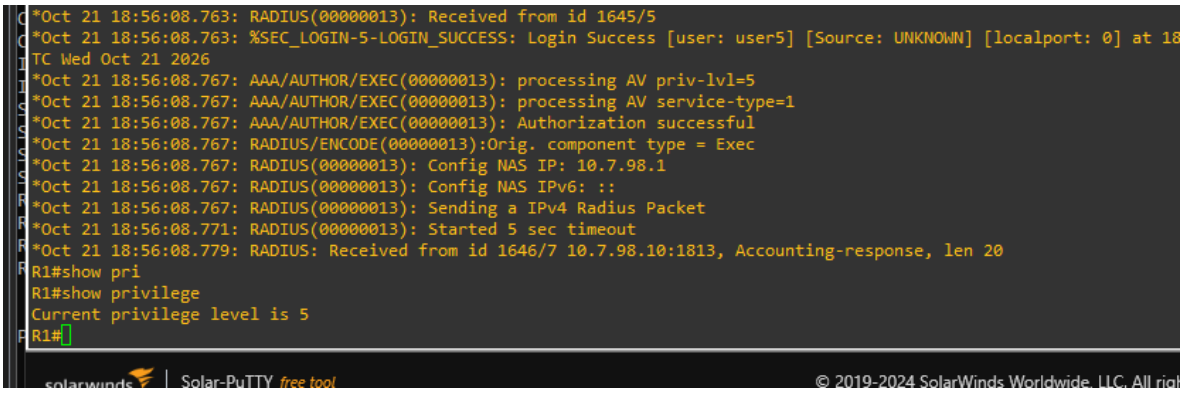
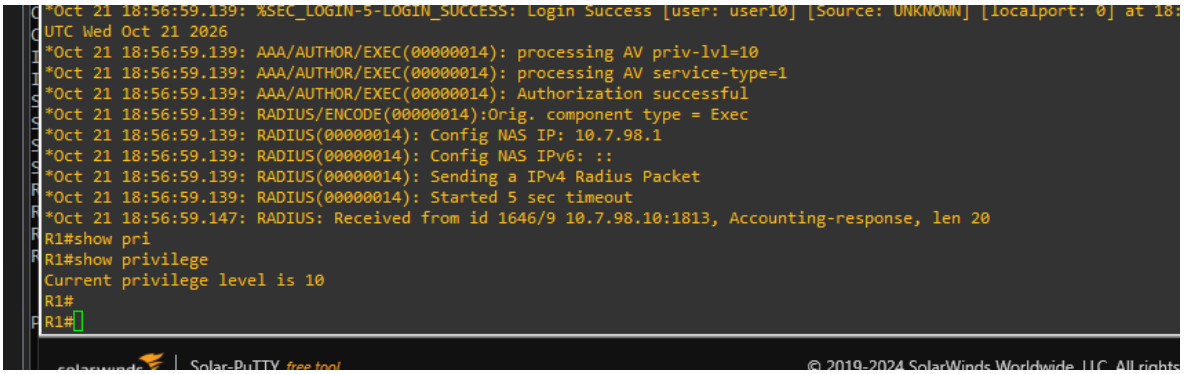
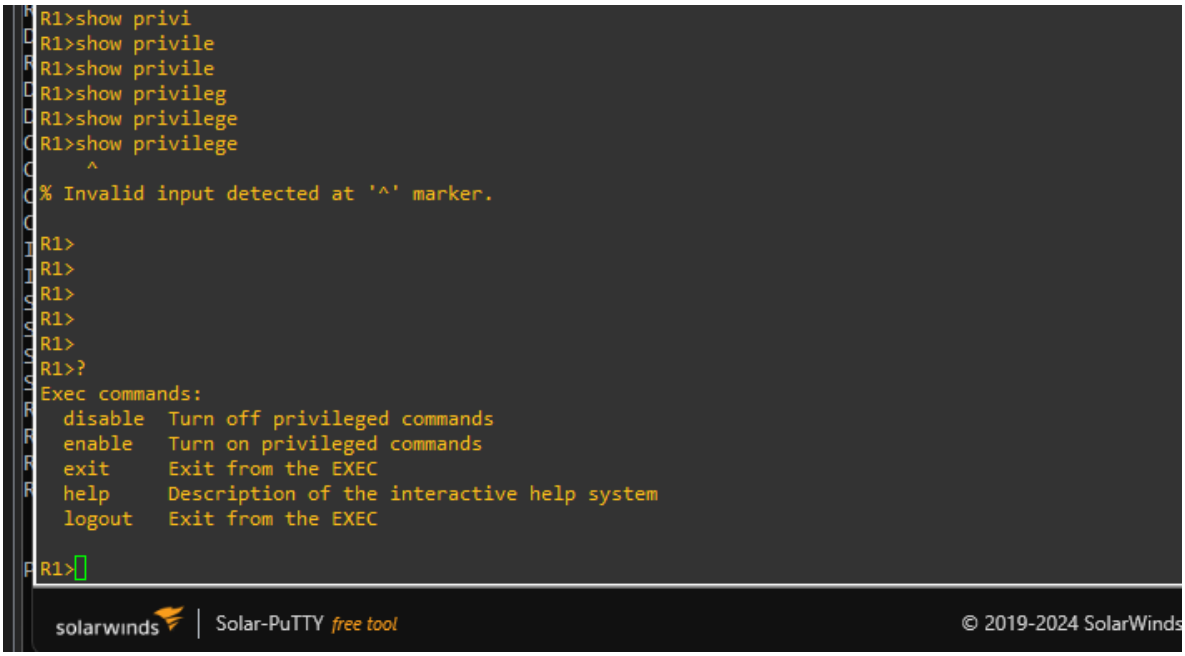

*Oct 21 18:56:08.763: RADIUS(00000013): Received from id 1645/5
*Oct 21 18:56:08.763: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: user5] [Source: UNKNOWN] [localport: 0] at 18
TC Wed Oct 21 2026
*Oct 21 18:56:08.767: AAA/AUTHOR/EXEC(00000013): processing AV priv-lvl=5
*Oct 21 18:56:08.767: AAA/AUTHOR/EXEC(00000013): processing AV service-type=1
*Oct 21 18:56:08.767: AAA/AUTHOR/EXEC(00000013): Authorization successful
*Oct 21 18:56:08.767: RADIUS/ENCODE(00000013):Orig. component type = Exec
*Oct 21 18:56:08.767: RADIUS(00000013): Config NAS IP: 10.7.98.1
*Oct 21 18:56:08.767: RADIUS(00000013): Config NAS IPv6: ::
*Oct 21 18:56:08.767: RADIUS(00000013): Sending a IPv4 Radius Packet
*Oct 21 18:56:08.771: RADIUS(00000013): Started 5 sec timeout
*Oct 21 18:56:08.779: RADIUS: Received from id 1646/7 10.7.98.10:1813, Accounting-response, len 20
R1#show pri
R1#show privilege
Current privilege level is 5
R1#
solarwinds | Solar-PuTTY free tool
© 2019-2024 SolarWinds Worldwide, LLC. All rig


*Oct 21 18:56:59.139: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: user10] [Source: UNKNOWN] [localport: 0] at 18:
UTC Wed Oct 21 2026
*Oct 21 18:56:59.139: AAA/AUTHOR/EXEC(00000014): processing AV priv-lvl=10
*Oct 21 18:56:59.139: AAA/AUTHOR/EXEC(00000014): processing AV service-type=1
*Oct 21 18:56:59.139: AAA/AUTHOR/EXEC(00000014): Authorization successful
*Oct 21 18:56:59.139: RADIUS/ENCODE(00000014):Orig. component type = Exec
*Oct 21 18:56:59.139: RADIUS(00000014): Config NAS IP: 10.7.98.1
*Oct 21 18:56:59.139: RADIUS(00000014): Config NAS IPv6: ::
*Oct 21 18:56:59.139: RADIUS(00000014): Sending a IPv4 Radius Packet
*Oct 21 18:56:59.139: RADIUS(00000014): Started 5 sec timeout
*Oct 21 18:56:59.147: RADIUS: Received from id 1646/9 10.7.98.10:1813, Accounting-response, len 20
R1#show pri
R1#show privilege
Current privilege level is 10
R1#
R1#
© 2019-2024 SolarWinds Worldwide, LLC. All rights


R1>show privi
R1>show privile
R1>show privile
R1>show privileg
R1>show privilege
R1>show privilege
^
% Invalid input detected at '^' marker.
R1>
R1>
R1>
R1>
R1>
R1>?
Exec commands:
disable Turn off privileged commands
enable Turn on privileged commands
exit Exit from the EXEC
help Description of the interactive help system
logout Exit from the EXEC
R1>
solarwinds | Solar-PuTTY free tool
© 2019-2024 SolarWinds

```
Supplemental End User License Restrictions

This IOSv software is provided AS-IS without warranty of any kind. Under no circumstances may this software be used sepa
 from the Cisco Modeling Labs Software that this software was provided with, or deployed or used as part of a production
ironment.

By using the software, you agree to abide by the terms and conditions of the Cisco End User License Agreement at http://
cisco.com/go/eula. Unauthorized use or distribution of this software is expressly prohibited.

SW1#show
*Oct 21 19:18:28.552: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: user10] [Source: UNKNOWN] [localport: 0] at 19:18
UTC Wed Oct 21 2026
SW1#show pr
SW1#show pr
SW1#show prr
SW1#show pri
SW1#show privilege
Current privilege level is 10
SW1#
```

```
Username:
Username: user5
Password:

IOSv - Cisco Systems Confidential -

Supplemental End User License Restrictions

This IOSv software is provided AS-IS without warranty of any kind. Under no circumstances may this software b
 from the Cisco Modeling Labs Software that this software was provided with, or deployed or used as part of a
ironment.

By using the software, you agree to abide by the terms and conditions of the Cisco End User License Agreement
cisco.com/go/eula. Unauthorized use or distribution of this software is expressly prohibited.

SW1#show
SW1#show
*Oct 21 19:18:59.760: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: user5] [Source: UNKNOWN] [localport: 0
TC Wed Oct 21 2026
SW1#show pr
SW1#show pri
SW1#show pri
SW1#show privil
SW1#show privile
SW1#show privilehe
SW1#show privile
SW1#?
Exec commands:
  access-enable    Create a temporary Access-List entry
  access-profile   Apply user-profile to interface
  clear            Reset functions
  connect          Open a terminal connection
  crypto           Encryption related commands.
  disable          Turn off privileged commands
  disconnect       Disconnect an existing network connection
  do-exec          Mode-independent "do-exec" prefix support
```

```
SW1#show privilege
Current privilege level is 15
SW1#exit
Connection to 10.7.98.2 closed.
❯ ssh -o KexAlgorithms=+diffie-hellman-group1-sha1,diffie-hellman-group14-sha1 \
-o HostKeyAlgorithms=+ssh-rsa \
-c aes128-ctr \
-l admin 10.7.98.2

IOSv - Cisco Systems Confidential -

Supplemental End User License Restrictions

This IOSv software is provided AS-IS without warranty of any kind. Under no circumstances may this soft
ware be used separate from the Cisco Modeling Labs Software that this software was provided with, or de
ployed or used as part of a production environment.

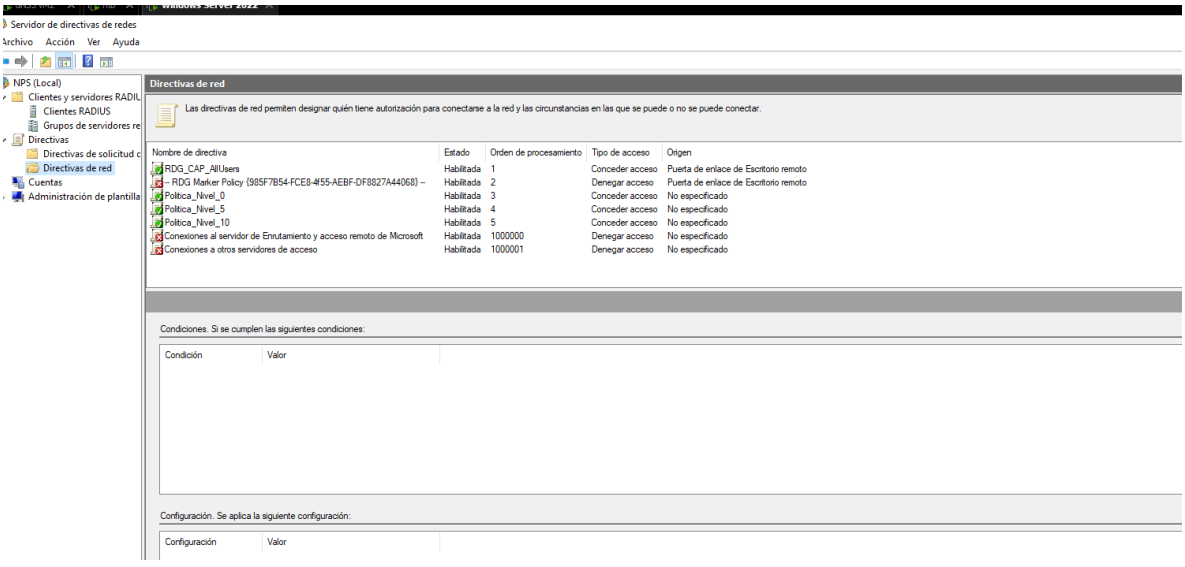
By using the software, you agree to abide by the terms and conditions of the Cisco End User License Agr
eement at http://www.cisco.com/go/eula. Unauthorized use or distribution of this software is expressly
prohibited.
(admin@10.7.98.2) Password:
ACCESO SOLO AUTORIZADO - Toda actividad es registrada
IOSv - Cisco Systems Confidential -

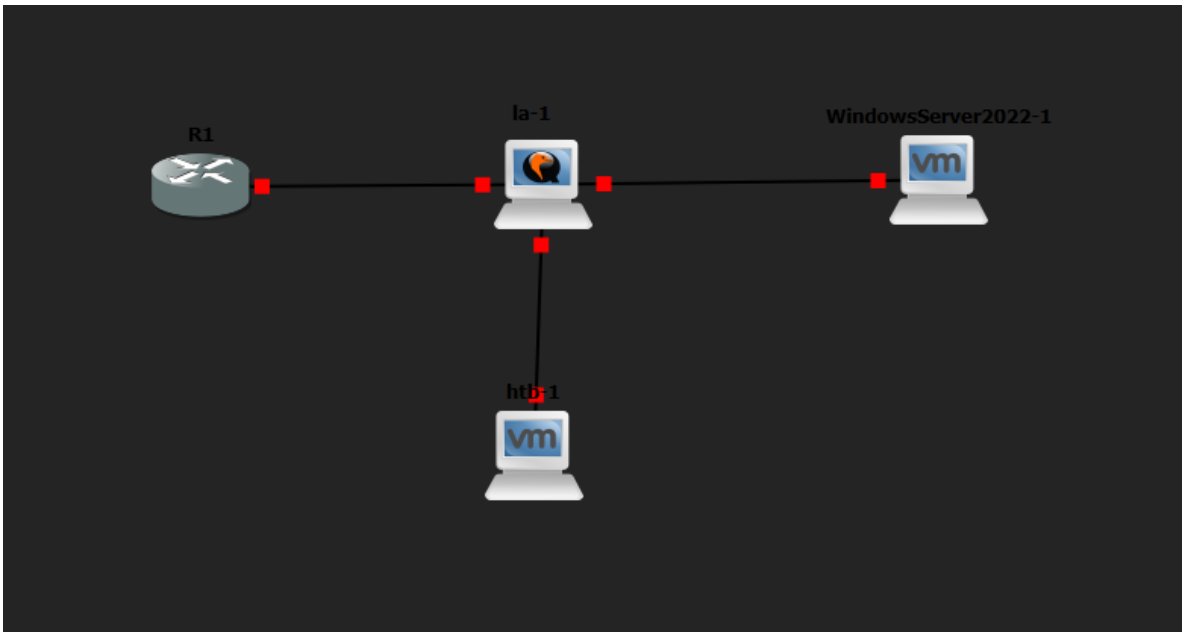
Supplemental End User License Restrictions

This IOSv software is provided AS-IS without warranty of any kind. Under no circumstances may this soft
ware be used separate from the Cisco Modeling Labs Software that this software was provided with, or de
ployed or used as part of a production environment.

By using the software, you agree to abide by the terms and conditions of the Cisco End User License Agr
eement at http://www.cisco.com/go/eula. Unauthorized use or distribution of this software is expressly
prohibited.
SW1#
```

```
prohibited.
SW1#
SW1#exit
Connection to 10.7.98.2 closed.
❯ ssh -o KexAlgorithms=+diffie-hellman-group1-sha1,diffie-hellman-group14-sha1 \
-o HostKeyAlgorithms=+ssh-rsa \
-c aes128-cbc \
-l admin 10.7.98.1
(admin@10.7.98.1) Password:
ACCESO SOLO AUTORIZADO - Toda actividad es registradaR1#
```




R1
la-1
WindowsServer2022-1
htb-1

```
PS C:\Users\Administrador>
PS C:\Users\Administrador> Get-DnsClientServerAddress -AddressFamily IPv4

InterfaceAlias               Interface Address ServerAddresses
                             Index     Family
--------------               --------- ------- --------------
Ethernet0 2                          9 IPv4    {10.7.98.10, 127.0.0.1}
Loopback Pseudo-Interface 1          1 IPv4    {}


PS C:\Users\Administrador> ping 10.7.98.1

Haciendo ping a 10.7.98.1 con 32 bytes de datos:
Respuesta desde 10.7.98.1: bytes=32 tiempo=10ms TTL=255
Respuesta desde 10.7.98.1: bytes=32 tiempo=11ms TTL=255
Respuesta desde 10.7.98.1: bytes=32 tiempo=3ms TTL=255
Respuesta desde 10.7.98.1: bytes=32 tiempo=9ms TTL=255

Estadísticas de ping para 10.7.98.1:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 3ms, Máximo = 11ms, Media = 8ms
PS C:\Users\Administrador> ping 10.7.98.2

Haciendo ping a 10.7.98.2 con 32 bytes de datos:
Respuesta desde 10.7.98.2: bytes=32 tiempo=4ms TTL=255
Respuesta desde 10.7.98.2: bytes=32 tiempo=5ms TTL=255
Respuesta desde 10.7.98.2: bytes=32 tiempo=2ms TTL=255
Respuesta desde 10.7.98.2: bytes=32 tiempo=2ms TTL=255

Estadísticas de ping para 10.7.98.2:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 2ms, Máximo = 5ms, Media = 3ms
PS C:\Users\Administrador> Get-NetUDPEndpoint |
>> Where-Object {$_.LocalPort -in 1812,1813} |
>> Format-Table LocalAddress,LocalPort,OwningProcess

LocalAddress               LocalPort OwningProcess
------------               --------- -------------
fe80::cd54:f4ce:c75e:d73%9      1813          2520
::ffff:10.7.98.10               1813          2520
fe80::cd54:f4ce:c75e:d73%9      1812          2520
::ffff:10.7.98.10               1812          2520


PS C:\Users\Administrador>
```

```
Name                  : SW1
Address               : 10.7.98.2
AuthAttributeRequired : False
SharedSecret          : Cisco1234
VendorName            : Cisco
Enabled               : True



PS C:\Users\Administrador> Get-NpsRadiusClient | Select-Object Name,Address,VendorName,Enabled

Name Address   VendorName Enabled
---- -------   ---------- -------
R1   10.7.98.1 Cisco         True
SW1  10.7.98.2 Cisco         True


PS C:\Users\Administrador> Get-NetIPConfiguration


InterfaceAlias       : Ethernet0 2
InterfaceIndex       : 9
InterfaceDescription : Intel(R) PRO/1000 MT Network Connection
NetProfile.Name      : Red 4
IPv4Address          : 10.7.98.10
IPv6DefaultGateway   :
IPv4DefaultGateway   : 10.7.98.1
DNSServer            : 10.7.98.10
                       127.0.0.1



PS C:\Users\Administrador> Get-NetIPAddress -AddressFamily IPv4 |
>> Where-Object {$_.IPAddress -eq "10.7.98.10"}


IPAddress         : 10.7.98.10
InterfaceIndex    : 9
InterfaceAlias    : Ethernet0 2
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 24
PrefixOrigin      : Manual
SuffixOrigin      : Manual
AddressState      : Preferred
ValidLifetime     : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource      : False
PolicyStore       : ActiveStore



PS C:\Users\Administrador>
```

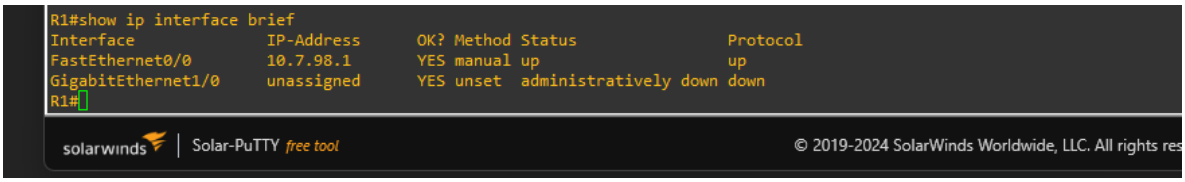
```
PS C:\Users\Administrador> Get-ADDomain | Select-Object DNSRoot,NetBIOSName,DomainMode

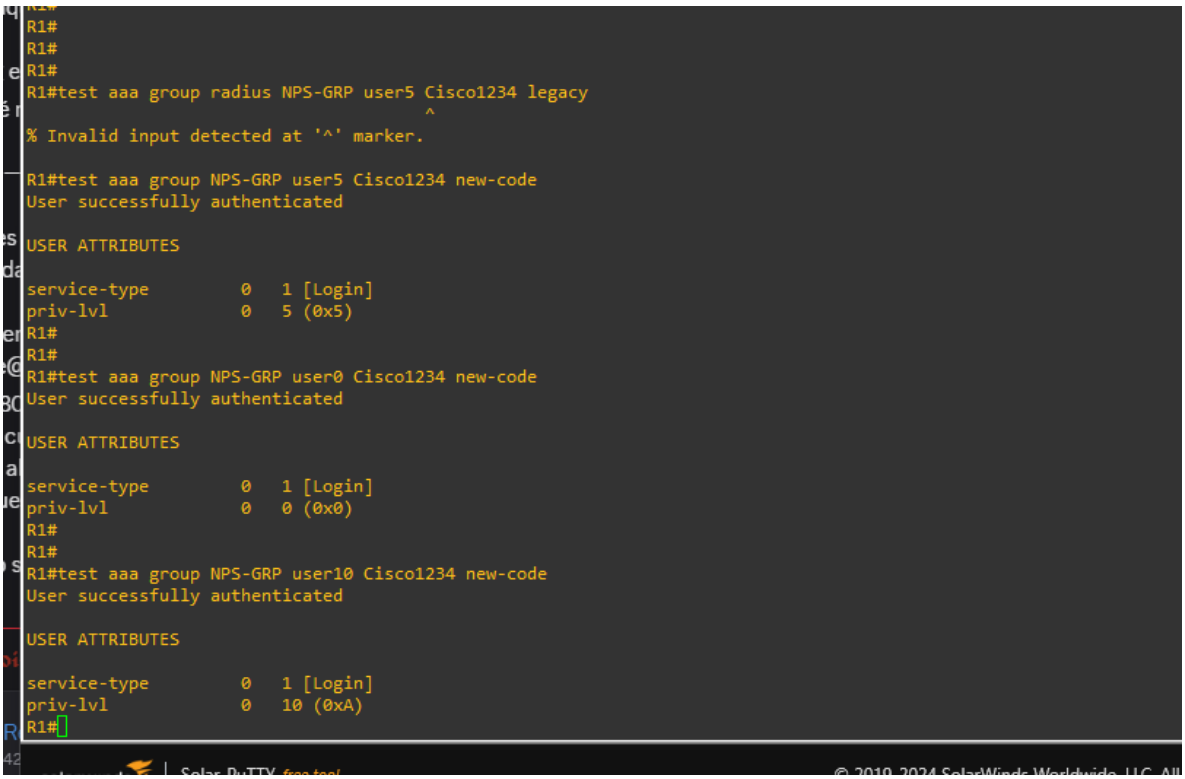
DNSRoot   NetBIOSName        DomainMode
-------   -----------        ----------
lab.local LAB         Windows2016Domain


PS C:\Users\Administrador> Get-NetFirewallRule |
>> Where-Object {$_.DisplayName -match "RADIUS|1812|1813|ICMP"} |
>> Format-Table DisplayName,Enabled,Direction,Action -AutoSize

DisplayName                                                                                                  Enabled Direction Action
-----------                                                                                                  ------- --------- ------
Redes principales: paquete demasiado grande (ICMPv6 de salida)                                                  True  Outbound  Allow
Redes principales: problema de parámetro (ICMPv6 de salida)                                                     True  Outbound  Allow
Redes principales: anuncio de enrutador (ICMPv6 de entrada)                                                     True   Inbound  Allow
Redes principales: destino inaccesible fragment. necesaria (ICMPv4 de entrada)                                  True   Inbound  Allow
Archivos e impresoras compartidos (petición eco: ICMPv4 de salida)                                              True  Outbound  Allow
Redes principales: solicitud de enrutador (ICMPv6 de salida)                                                    True  Outbound  Allow
Redes principales: solicitud de detección de vecinos (ICMPv6 de entrada)                                        True   Inbound  Allow
Redes principales: informe de escucha de multidifusión v2 (ICMPv6 de salida)                                    True  Outbound  Allow
Diagnósticos de redes principales: solicitud de eco ICMP (ICMPv6 de salida)                                    False  Outbound  Allow
Diagnóstico de redes principales: solicitud de eco ICMP (ICMPv4 de salida)                                     False  Outbound  Allow
Redes principales: escucha de multidifusión finalizada (ICMPv6 de salida)                                       True  Outbound  Allow
Redes principales: solicitud de enrutador (ICMPv6 de entrada)                                                   True   Inbound  Allow
Archivos e impresoras compartidos (petición eco: ICMPv6 de salida)                                              True  Outbound  Allow
Redes principales: anuncio de enrutador (ICMPv6 de salida)                                                      True  Outbound  Allow
Redes principales: anuncio de detección de vecinos (ICMPv6 de entrada)                                          True   Inbound  Allow
Redes principales: problema de parámetro (ICMPv6 de entrada)                                                    True   Inbound  Allow
Archivos e impresoras compartidos (petición eco: ICMPv4 de entrada)                                             True   Inbound  Allow
Redes principales: anuncio de detección de vecinos (ICMPv6 de salida)                                           True  Outbound  Allow
Redes principales: consulta de escucha de multidifusión (ICMPv6 de salida)                                      True  Outbound  Allow
Diagnósticos de redes principales: solicitud de eco ICMP (ICMPv6 de entrada)                                   False   Inbound  Allow
Redes principales: tiempo superado (ICMPv6 de salida)                                                           True  Outbound  Allow
Supervisión de máquina virtual (Solicitud de eco - ICMPv4 de entrada)                                          False   Inbound  Allow
Supervisión de máquina virtual (Solicitud de eco - ICMPv6 de entrada)                                          False   Inbound  Allow
Redes principales: informe de escucha de multidifusión (ICMPv6 de salida)                                       True  Outbound  Allow
Redes principales: paquete demasiado grande (ICMPv6 de entrada)                                                 True   Inbound  Allow
Redes principales: escucha de multidifusión finalizada (ICMPv6 de entrada)                                      True   Inbound  Allow
Redes principales: destino inaccesible (ICMPv6 de entrada)                                                      True   Inbound  Allow
Archivos e impresoras compartidos (petición eco: ICMPv6 de entrada)                                             True   Inbound  Allow
Diagnóstico de redes principales: solicitud de eco ICMP (ICMPv4 de salida)                                     False  Outbound  Allow
Redes principales: solicitud de detección de vecinos (ICMPv6 de salida)                                         True  Outbound  Allow
Diagnósticos de redes principales: solicitud de eco ICMP (ICMPv6 de salida)                                    False  Outbound  Allow
Redes principales: informe de escucha de multidifusión v2 (ICMPv6 de entrada)                                   True   Inbound  Allow
Redes principales: consulta de escucha de multidifusión (ICMPv6 de entrada)                                     True   Inbound  Allow
Redes principales: informe de escucha de multidifusión (ICMPv6 de entrada)                                      True   Inbound  Allow
Diagnóstico de redes principales: solicitud de eco ICMP (ICMPv4 de entrada)                                    False   Inbound  Allow
Redes principales: tiempo superado (ICMPv6 de entrada)                                                          True   Inbound  Allow
Diagnóstico de redes principales: solicitud de eco ICMP (ICMPv4 de entrada)                                    False   Inbound  Allow
Diagnósticos de redes principales: solicitud de eco ICMP (ICMPv6 de entrada)                                   False   Inbound  Allow
Controlador de dominio de Active Directory: solicitud de eco (ICMPv4 de salida)                                 True  Outbound  Allow
Controlador de dominio de Active Directory: solicitud de eco (ICMPv6 de salida)                                 True  Outbound  Allow
Redes principales: anuncio de detección de vecinos (ICMPv6 de salida)                                           True  Outbound  Allow
Redes principales: consulta de escucha de multidifusión (ICMPv6 de salida)                                      True  Outbound  Allow
Diagnósticos de redes principales: solicitud de eco ICMP (ICMPv6 de entrada)                                   False   Inbound  Allow
Redes principales: tiempo superado (ICMPv6 de salida)                                                           True  Outbound  Allow
Supervisión de máquina virtual (Solicitud de eco - ICMPv4 de entrada)                                          False   Inbound  Allow
Supervisión de máquina virtual (Solicitud de eco - ICMPv6 de entrada)                                          False   Inbound  Allow
Redes principales: informe de escucha de multidifusión (ICMPv6 de salida)                                       True  Outbound  Allow
Redes principales: paquete demasiado grande (ICMPv6 de entrada)                                                 True   Inbound  Allow
Redes principales: escucha de multidifusión finalizada (ICMPv6 de entrada)                                      True   Inbound  Allow
Redes principales: destino inaccesible (ICMPv6 de entrada)                                                      True   Inbound  Allow
Archivos e impresoras compartidos (petición eco: ICMPv6 de entrada)                                             True   Inbound  Allow
Diagnóstico de redes principales: solicitud de eco ICMP (ICMPv4 de salida)                                     False  Outbound  Allow
Redes principales: solicitud de detección de vecinos (ICMPv6 de salida)                                         True  Outbound  Allow
Diagnósticos de redes principales: solicitud de eco ICMP (ICMPv6 de salida)                                    False  Outbound  Allow
Redes principales: informe de escucha de multidifusión v2 (ICMPv6 de entrada)                                   True   Inbound  Allow
Redes principales: consulta de escucha de multidifusión (ICMPv6 de entrada)                                     True   Inbound  Allow
Redes principales: informe de escucha de multidifusión (ICMPv6 de entrada)                                      True   Inbound  Allow
Diagnóstico de redes principales: solicitud de eco ICMP (ICMPv4 de entrada)                                    False   Inbound  Allow
Redes principales: tiempo superado (ICMPv6 de entrada)                                                          True   Inbound  Allow
Diagnóstico de redes principales: solicitud de eco ICMP (ICMPv4 de entrada)                                    False   Inbound  Allow
Diagnósticos de redes principales: solicitud de eco ICMP (ICMPv6 de entrada)                                   False   Inbound  Allow
Controlador de dominio de Active Directory: solicitud de eco (ICMPv4 de salida)                                 True  Outbound  Allow
Controlador de dominio de Active Directory: solicitud de eco (ICMPv6 de salida)                                 True  Outbound  Allow
Controlador de dominio de Active Directory: solicitud de eco (ICMPv4 de entrada)                                True   Inbound  Allow
Controlador de dominio de Active Directory: solicitud de eco (ICMPv6 de entrada)                                True   Inbound  Allow
ICMP entrada                                                                                                    True   Inbound  Allow
ICMP desde VPN                                                                                                  True   Inbound  Allow
Servidor de directivas de redes (autenticación RADIUS heredada: UDP de entrada)                                 True   Inbound  Allow
Servidor de directivas de redes (administración de cuentas RADIUS heredada: UDP de entrada)                     True   Inbound  Allow
Servidor de directivas de redes (autenticación RADIUS: UDP de entrada)                                          True   Inbound  Allow
Servidor de directivas de redes (contabilización de cuentas RADIUS: UDP de entrada)                             True   Inbound  Allow
RADIUS Auth (UDP 1812)                                                                                          True   Inbound  Allow
RADIUS Acct (UDP 1813)                                                                                          True   Inbound  Allow
RADIUS-Auth-In                                                                                                  True   Inbound  Allow
RADIUS-Acct-In                                                                                                  True   Inbound  Allow


PS C:\Users\Administrador> _
```


R1#show ip interface brief
Interface   IP-Address   OK? Method Status   Protocol
FastEthernet0/0   10.7.98.1   YES manual up   up
GigabitEthernet1/0   unassigned   YES unset administratively down down
R1#
solarwinds | Solar-PuTTY free tool
© 2019-2024 SolarWinds Worldwide, LLC. All rights res


R1#
R1#
R1#
R1#test aaa group radius NPS-GRP user5 Cisco1234 legacy
                                  ^
% Invalid input detected at '^' marker.
R1#test aaa group NPS-GRP user5 Cisco1234 new-code
User successfully authenticated
USER ATTRIBUTES
service-type   0   1 [Login]
priv-lvl   0   5 (0x5)
R1#
R1#
R1#test aaa group NPS-GRP user0 Cisco1234 new-code
User successfully authenticated
USER ATTRIBUTES
service-type   0   1 [Login]
priv-lvl   0   0 (0x0)
R1#
R1#
R1#test aaa group NPS-GRP user10 Cisco1234 new-code
User successfully authenticated
USER ATTRIBUTES
service-type   0   1 [Login]
priv-lvl   0   10 (0xA)
R1#